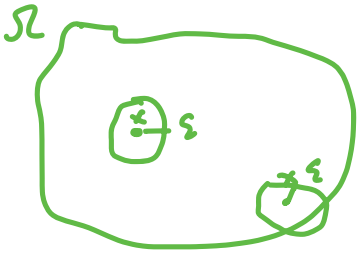
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Approximation of bad func by good func

Def 3 $\forall u \in L^1_{loc}(\Omega)$, the regularization u_ε is

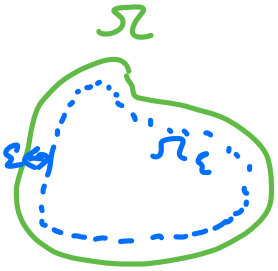
$$u_\varepsilon(x) = \int_\Omega j_\varepsilon(x-y)u(y)\,dy, \quad j_\varepsilon(x) = \frac{1}{\varepsilon^n} j\left(\frac{x}{\varepsilon}\right)$$

- $\forall x \in \Omega$, $u_\varepsilon(x)$ is well-defined for $0 < \varepsilon < \operatorname{dist}(x, \partial\Omega)$



- For fixed ε, $u_\varepsilon(x)$ is well-defined for $x \in \Omega_\varepsilon$

$$\Omega_\varepsilon = \{x \in \Omega : \operatorname{dist}(x, \partial\Omega) > \varepsilon\}$$



- If $u \in L^1(\Omega)$, then $u_\varepsilon(x)$ is well-defined on $\mathbb{R}^n$.
- If $u \in L^1(\Omega)$, $u \equiv 0$ in Ω^c then $u_\varepsilon(x) = \int_{\mathbb{R}^n} j_\varepsilon(x-y)u(y)\,dy = j_\varepsilon * u$.

Lemma 1 Suppose $u \in L^1_{loc}(\Omega)$. Then $\forall$ fixed small $\varepsilon > 0$, $u_\varepsilon \in C^\infty(\Omega_\varepsilon)$

Moreover, if $u \in L^p(\Omega)$ $(1 \le p \le \infty)$, then $u_\varepsilon \in C^\infty(\mathbb{R}^n)$ & $u_\varepsilon \in C_c^\infty(\mathbb{R}^n)$ provided Ω bdd.

pf: $\forall$ fixed $x \in \Omega_\varepsilon$, $\exists\, \delta > \varepsilon$ s.t. $\overline{B_\delta(x)} \subset \Omega$.

Observe: if z is close to x, then $B_\varepsilon(z) \subset B_\delta(x)$. Let $e_i := (0, \cdots \overset{i\text{th.}}{\underset{\uparrow}{1}}, \cdots 0)$

If $s \in \mathbb{R}$ small, then $B_\varepsilon(x + se_i) \subset B_\delta(x)$, consider

$$\frac{u_\varepsilon(x+he_i) - u_\varepsilon(x)}{h} = \int_\Omega \frac{j(x+he_i-y) - j(x-y)}{h} u(y)\,dy = \int_\Omega \frac{\partial}{\partial x_i} j_\varepsilon(x+se_i-y)u(y)\,dy \qquad (0<s<h)$$

By DCT,

$$\left|\frac{\partial}{\partial x_i} j_\varepsilon(x+se_i-y)u(y)\right| \le \left\|\frac{\partial}{\partial x_i} j_\varepsilon\right\|_{L^\infty} |u(y)| \mathbb{1}_{B_\varepsilon(x)} \in L^1(\Omega)$$

$$\Rightarrow \lim_{h \to 0^+} \frac{u_\varepsilon(x+he_i) - u_\varepsilon(x)}{h} = \lim_{h \to 0^+} \int_\Omega \frac{\partial}{\partial x_i} j_\varepsilon(x+se_i-y)u(y)\,dx$$

$$\frac{\partial}{\partial x_i} u_\varepsilon(x) = \int_\Omega \frac{\partial}{\partial x_i} j_\varepsilon(x-y)u(y)\,dx.$$

Continuity of $\frac{\partial}{\partial x_i} u_\varepsilon(x)$: $\{x_k\}_{k=1}^{\infty} \subset \Omega_\varepsilon$, $x_k \to x$ $(k\to\infty)$, $x \in \Omega_\varepsilon$

$$\left|\frac{\partial u_\varepsilon}{\partial x_i}(x_k) - \frac{\partial u_\varepsilon}{\partial x_i}(x)\right| = \left|\int_\Omega \left[\frac{\partial}{\partial x_i} j_\varepsilon(x_k - y) - \frac{\partial}{\partial x_i} j_\varepsilon(x-y)\right] u(y)\,dy\right|$$

By DCT,
$$\leq \int_\Omega \left|\frac{\partial}{\partial x_i} j_\varepsilon(x_k-y) - \frac{\partial}{\partial x_i} j_\varepsilon(x-y)\right| |u(y)|\,dy \to 0 \quad (k\to\infty).$$

$$\left|\frac{\partial}{\partial x_i} j_\varepsilon(x_k-y) - \frac{\partial}{\partial x_i} j_\varepsilon(x-y)\right| |u(y)| \leq 2 \left\|\frac{\partial}{\partial x_i} j_\varepsilon\right\|_{L^\infty(\mathbb{R}^n)} |u(y)| \mathbb{1}_{B_\delta(x)} \in L^1(\Omega)$$

$$\Rightarrow \frac{\partial u_\varepsilon}{\partial x_i}(x_k) \to \frac{\partial u_\varepsilon}{\partial x_i}(x) \quad (k\to\infty)$$

Similarly, $u_\varepsilon \in C^K(\Omega)$ $\forall K \geq 1$, $K \in \mathbb{N}$.

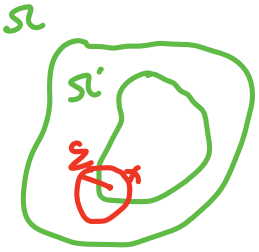
Moreover: Suppose $u \in L^1(\Omega)$, u_ε is well-defined in $\mathbb{R}^n$.

Simple and similar proof $\Rightarrow u_\varepsilon \in C^\infty(\Omega)$.

Lemma2 If $u \in C(\Omega)$ $(L^1_{loc}(\Omega))$, then $\forall \Omega' \subset\subset \Omega$ $(\Omega' \subset \overline{\Omega'} \subset \Omega)$

$u_\varepsilon \to u$ $(\varepsilon \to 0^+)$ in $C(\overline{\Omega'})$ (i.e. $u_\varepsilon \rightrightarrows u$ in $\overline{\Omega'}$)

Pf:



Let $\varepsilon \in (0, \operatorname{dist}(\overline{\Omega'}, \partial\Omega))$. Then

$u_\varepsilon(x) = \int_\Omega j_\varepsilon(x-y) u(y)\,dy$ is well-defined on $\overline{\Omega'}$

$$\overset{z=\frac{x-y}{\varepsilon}}{=} \int_{B_{1(0)}} j_\varepsilon(z)\, u(x-\varepsilon z)\,dz \qquad u(x) = \int_{B_{1(0)}} j_\varepsilon(z)\, u(x)\,dz$$

W.T.S: $\forall \delta > 0$, $\exists \sigma > 0$, s.t. if $\varepsilon < \sigma$, $|u_\varepsilon(x) - u(x)| < \delta$ $\forall x \in \overline{\Omega'}$

Take a Ω'', $\Omega' \subset\subset \Omega'' \subset\subset \Omega$. Then for all small $\varepsilon > 0$ & $\forall z \in B_{1(0)}$, $\forall x \in \overline{\Omega'}$, x, $x - z\varepsilon \in \overline{\Omega''}$. Since $u \in C^0(\overline{\Omega''})$, $\overline{\Omega''}$ is cpt $\Rightarrow u$ is uniformly continuous on $\overline{\Omega''}$.

$$|u_\varepsilon(x) - u(x)| = \left|\int_{B_{1(0)}} j_\varepsilon(z)(u(x-\varepsilon z) - u(x))\,dz\right| \leq \int_{B_{1(0)}} j_\varepsilon(z) |u(x-\varepsilon z) - u(x)|\,dz$$

$$\leq \delta \int_{B_{1(0)}} j_\varepsilon(z)\,dz = \delta$$

Lemma3 $u \in L^1_{loc}(\Omega)$, $u_\varepsilon \to u$ a.e. as $\varepsilon \to 0$

Pf: By Lebesgue's Differentiation Thm: $f: \mathbb{R}^n \to \mathbb{R}$, $f \in L^1_{loc}(\mathbb{R}^n)$

$\Rightarrow$ (1) For a.e. $x_0 \in \mathbb{R}^n$, $\fint_{B_r(x_0)} f(x)\,dx \longrightarrow f(x_0)$ $(r \to 0^+)$

(2) For a.e. $x_0 \in \mathbb{R}^n$ (Leb point), $\fint_{B_r(x_0)} |f(x)-f(x_0)|dx \to 0$ $(r \to 0^+)$

f is approximately continuous at x_0.

For a Leb point x of u, $|u_\varepsilon(x)-u(x)| \le \int_{B_\varepsilon(x)} j_\varepsilon(x-y)|u(y)-u(x)|dy$

$\le C \fint_{B_\varepsilon(x)} |u(y)-u(x)|dy \to 0$ $(\varepsilon \to 0^+)$

Lemma4 $1 \le p < \infty$, $u \in L^p_{loc}(\Omega)$ or $L^p(\Omega)$. Then $u_\varepsilon \xrightarrow{\varepsilon \to 0^+} u$ in $L^p_{loc}(\Omega)$ or L^p

pf: Suppose $u \in L^p_{loc}(\Omega)$, W.T.S: $\Omega' \subset\subset \Omega$, $\int_{\Omega'} |u_\varepsilon - u|^p dx \xrightarrow{\varepsilon \to 0^+} 0$

$\|u_\varepsilon(x)-u(x)\|_{L^p(\Omega')} = \| \int_{B_1(0)} j(z)(u(x-\varepsilon z)-u(x))dz \|_{L^p}$. By Minkowski's ineq.

$\le \int_{B_1(0)} j(z) \|u(x-\varepsilon z)-u(x)\|_{L^p} dz$ ···(*)

Take Ω'': $\Omega' \subset\subset \Omega'' \subset\subset \Omega$, $\hat{u} \triangleq \begin{cases} u, & x \in \Omega'' \\ 0, & x \in \Omega''^c \end{cases}$ $\hat{u} \in L^1(\mathbb{R}^n)$

$\|u(x-\varepsilon z)-u(x)\|_{L^p(\Omega')} \le \|u(x-\varepsilon z)-u(x)\|_{L^p(\Omega'')} \xrightarrow{\varepsilon \to 0^+} 0$ uniformly for z

Continuity of L^p-norm: $\|f(\cdot+h)-f(\cdot)\|_{L^p} \to 0$ $(h \to 0)$ $p < \infty$

(*) By DCT, $\le j_\varepsilon(z) 2 \|\hat{u}\|_{L^p(\mathbb{R}^n)}$, $(*) \to 0$ $(\varepsilon \to 0^+)$